ESE 4025 One-Shot Notes

HVAC&R System Design · KUET · past papers 2020, 2021, 2023, 2024 · answers from the course slides

Section A: Refrigeration

Units and basics

Refrigeration: removing heat from a substance under controlled conditions and keeping it below the surroundings' temperature.

1 TR = melting 1 tonne ice from and at 0 °C in 24 h = 232.6 kJ/min; practice: **210 kJ/min = 3.5 kW**.

```
COP = RE/W = (h1 - h4)/(h2 - h1); COP_HP
= COP_R + 1
Carnot COP_R = TL/(TH - TL)
```

Refrigerator: wants Q_L (cold side). Heat pump: wants Q_H (hot side). Refrigeration controls temperature only; AC controls temperature + humidity + air motion + purity.

Ice / product load recipe

```
Q = m[cΔT + latent] / time(min) ÷ 210 =
TR
ice latent 335 kJ/kg, water c = 4.187, 1
t ice/day = 3.88 kW
```

2000 kg, 0→45 °C: 3.46 TR · 1500 kg, 0→25 °C: 2.18 TR · 600 kg fruit 25→−5 °C in 8 h (c 1.255, L 105): 0.85 TR

VCR cycle (C-C-E-E)

1→2	Compressor, isentropic, w = h2 − h1
2→3	Condenser, const p, q = h2 − h3
3→4	Expansion valve, h4 = h3 (throttling)
4→1	Evaporator, RE = h1 − h4

Throttle replaces turbine; dry vapour compressed (Carnot impractical: wet compression, wet expansion).

Dry sat. after compression: s1 = s2 = s_{g2}; x1 = (s1 − s_{f1})/(s_{g1} − s_{f1}); h1 = h_{f1} + x1h_{fg1}.

Subcooling: RE ↑, W same → COP ↑ always.
Superheat: RE ↑ and W ↑; COP may go either way; protects compressor.

Effect of temperatures

T_{evap} ↑ → RE ↑, W ↓, v1 ↓ (capacity ↑) → COP ↑.

T_{cond} ↑ → RE ↓, W ↑, discharge temp ↑ → COP ↓.

Evaporator temp has more influence: ∂COP/∂Te = Tc/ΔT^2 > Te/ΔT^2. R22 example: +5 K evap → 4.13, −5 K cond → 4.11 (base 3.61).

R22 numerical recipe

```
find p_e, p_c; h1 (with superheat), v1,
s1
h2 at p_c with s2 = s1; h3 = h_f(Tc -
subcool) = h4
m = Q/(h1-h4); V = m·v1; P = m(h2-h1)
```

80 kW, −10/45 °C, 5 K sub, 10 K sup: h1 408.2, h2 450.4, h4 249.6, v1 0.0686 → m 0.505 kg/s, **V 0.0346 m³/s, COP 3.75, P 21.3 kW**.

2020 table (50 kW, 35/−10, h2 435.2): h1 401.06, h3 243.14 → RE 157.9, m 0.317, P 10.8 kW, COP 4.63.

Ammonia ice plant recipe

Q = tonnes × 3.88 kW. x1 from s, h1, COP_{th} = (h1 − h_{f3})/(h2 − h1), × %, P = Q/COP.

25 t, 15/−5, 45%: COP_{th} 12.8 → **16.9 kW**. 30 t, 20/−10, 55%: 8.15 → **26.0 kW**. Slide: 30 t, 25/−15, 60% → 33.2 kW.

Multistage

When: T_c − T_e > 40 K (Gosney), high pressure ratio.

Single stage then has: big cylinder, high discharge temp (oil burns), low η_{vol}, high friction/cost.

Advantages: less work, better η_{vol}, low discharge temp, more RE, less leakage, uniform torque, cheaper compressor. **Disadvantages:** cost, complexity.

```
p_i = √(p_c · p_e)
```

Flash cooler: LP comp 1→2, mix with flash gas → 3, HP comp → 4, condense → 5, valve 1 → 6 in flash chamber, vapour 7 to HP, liquid 8 → valve 2 → 9 evaporator. Flash gas skips evaporator and LP stage.

Single comp + multiple exp. valves: E3→E2→E1 in series, back-pressure valves, vapours mix to one compressor. **Individual compressors:** each evaporator has own compressor; COP higher.

Refrigerants

Criteria: thermodynamic, environmental & safety, economic.

Thermo: suction p above atm, low discharge p, low pressure ratio, high latent heat, low isentropic index, low liquid c_p, high vapour c_p, high k, low viscosity.

Environment: ODP = 0 (R11 = 1, Cl/Br destroy ozone), low GWP (CO_2 = 1), low TEWI (direct leakage + indirect energy CO_2).

Safety: non-toxic, non-flammable, chemically stable, compatible, oil miscible (NH_3 no, R12 yes, R22 partly), easy leak detection. ASHRAE A/B toxicity, 1/2/3 flammability.

Compressors

Classes: method (recip, rotary, centrifugal), strokes (single/double acting), stages, drive (direct/belt), prime mover (open, semi-hermetic, hermetic).

Hermetic: motor + compressor on one shaft in one welded shell; no shaft seal. +: no leakage, quiet, compact, simple lubrication. −: inaccessible, separate pump for charging, burnout contaminates.

Centrifugal vs recip +: no valves/pistons (long life), no vibration, quiet, 3000+ rpm direct drive, big volumes, 50-5000 TR, high efficiency, less floor area. −: surging <35% load, not below 50 TR.

Absorption (VAR)

Compressor replaced by absorber + pump + generator (heat input). Pairs: H_2O-LiBr (AC, >0 °C), NH_3-H_2O (below 0 °C).

	Single	Double	Triple
Generators	1	2 (HTG vapour heats LTG)	3
COP	0.7	1.0-1.2	1.5-1.7
Heat	80-120 °C	150-170 °C	200+ °C

VAR vs VCR: heat vs work; low vs high COP; few vs many moving parts; part load fine vs drops; quiet; bulky small units; good where waste heat is cheap.

Heat sources: steam, hot water, direct gas/oil firing, waste heat/exhaust, solar, geothermal.

LiBr problems: crystallization, air leakage (vacuum, purge), corrosion (inhibitors), freezing below ~4 °C, large vapour volume.

Practical plant performance

Full load: COP, EER, kW/ton. Part load: SEER, IPLV/NPLV.

```
EER = Btu/h ÷ W; COP = EER/3.412; kW/ton = 12/EER
IPLV = 0.01A + 0.42B + 0.45C + 0.12D (100/75/50/25%)
```

HSPF = heating season Btu ÷ Wh (heat pumps). Energy Star = EPA efficiency label.

Other short answers

Air cycle: compress → cool → expand in turbine → absorb; COP = $1/(r_p^{(\gamma-1)/\gamma} - 1)$. Aircraft (2020): Q = 50 × 1 × 35 = 1750 kJ/min = **8.33 TR**, COP ≈ 0.19.

TEV: bulb pressure opens; evaporator + spring pressure close; constant superheat.

Thermoelectric: Peltier, p-n couples; reverse current swaps hot and cold; COP 0.3-0.7. Magnetocaloric not commercial: costly magnets and Gd, small ΔT, low power density.

Cooling tower: natural/mechanical (forced, induced), counter/cross-flow. Range = in − out; Approach = out − WBT; ε = R/(R + A).

Linde: compress ~200 bar, HX regenerative cooling, J-T throttling, separate liquid; y = (h1 − h2)/(h1 − h_f).

Section B: Air conditioning

Psychrometric definitions

DBT	ordinary thermometer temperature
WBT	lowest temp by evaporating water; wet wick in moving air; DBT − WBT = depression
DPT	temp where vapour starts condensing on cooling at constant p (constant w)
w	kg vapour/kg dry air = 0.622 $p_v/(p_t - p_v)$
φ	p_v/p_{sat} at same DBT
μ	w/w_s at same T, p (0 dry, 1 saturated)
c_{pm}	1.005 + 1.88w ≈ 1.0216 kJ/kg K (heat to raise 1 kg d.a. + vapour by 1 K)
h	1.005t + w(2501 + 1.88t)
SHF	$Q_s/(Q_s + Q_L)$; 0.75-0.8 dry climate, 0.6 humid
BPF X	$(t_c - t_s)/(t_o - t_s)$; CF = 1 − X; lower with more rows, lower velocity, closer fins
ADP	effective coil surface temp; SHF line meets saturation curve

Saturated air: DBT = WBT = DPT. Always DBT ≥ WBT ≥ DPT.

SHF line: join chart reference point (26 °C, 50%) to SHF on protractor, draw parallel through room point. tanθ = (1/2451)(1 − SHF)/SHF.

Processes and air washer

Sensible heat/cool: w constant (coil above DPT). Cool + dehumidify: coil below DPT; $m_w = m_a(w_o - w_c)$, $Q = m_a(h_o - h_c)$. Heat + humidify: heating coil then steam/spray.

```
Qs = m·cpm·Δt   QL = m·2501·Δw   Qt = m·Δh
```

t_w	Air washer process
< DPT	cool + dehumidify (cool water)
DPT-WBT	cool + humidify (cool water)
= WBT	adiabatic/evaporative, h const (recirculate)
WBT-DBT	cool + humidify, h ↑ (heat water)
> DBT	heat + humidify (heat water)

Air washer = spray chamber + eliminators + pump + cooler/heater + make-up water. $\eta_H = 1 - X$.

Summer AC + ventilation + BPF recipe

```
mix: t1 = r·ti + (1-r)to (same for w, h)
coil: ts = ADP + X(t1 - ADP)
m_d = RSH / [1.02(ti - ts)]   m_o = (OA share)·m_d
OASH = m_o·1.02(to-ti)  OALH = m_o·2500(wo-wi)
GTH = RSH + RLH + m_o(ho - hi) ; TR = GTH/3.5
```

RSHF line: room i → supply. GSHF line: mixed 1 → coil ADP. ADP unknown? Try ADP until 1.02(ti − ts)/[2501(wi − ws)] = RSH/RLH.

2024: m_d 1.83 kg/s, OASH 8.4, OALH 8.0, **GTH 41.8 kW**. 2021 (OA 35 °C/65%): ADP 11.5 °C, 46.4 cmm, **91.1 kW**. Slide ex. (4:1, X 0.1): 1.73 kg/s, 37.7 kW.

Thermal comfort

Def: condition of mind expressing satisfaction with the thermal environment. **Factors:** air temp, MRT, RH, air velocity, activity (met), clothing (clo).

Comfort chart: ET lines + 60% RH top + 2 °C DPT bottom. Winter ET 20-23.5 °C, summer 22.5-26 °C, 1.2 met.

ET: temp of standard 50% RH environment giving the same skin heat loss. **Operative:** $(t_a + t_{mrt})/2$, radiation + convection.

```
PPD = 100 - 95 exp(-0.03353PMV&sup4; -
0.2179PMV²); PMV 0 → PPD 5%
```

So nobody can satisfy everyone. Indoor design: 24-26 °C, 50-60%. Outdoor design: statistical 1-2.5% weather values.

IAQ

SBS (symptoms vanish outside) vs BRI (real disease, persists). 90% time indoors; indoor air 2-5× worse.

Control: 1. Source control (low-VOC, intakes 8-10 ft high, 10-15 ft from exhausts, RH < 60%) 2. Filtration (MERV: panel 1-4, pleated 5-8, bag 9-15; HEPA 99.97% at 0.3 µm; ULPA 99.999% at 0.12 µm) 3. Dilution (ventilation, CO_2 sensor).

```
V = n·Np / (C_in - C_out); Np = 0.011 cfm
CO2; C_in 1000 ppm
1 person, 300 ppm out: 15.7 cfm
```

30 students: Khulna (400) **550 cfm = 0.26 m³/s**; Dhaka (800) **1650 cfm = 0.78 m³/s** (3×).

AC systems

Zone: space(s) controlled by one thermostat. **Split:** indoor evaporator + blower, outdoor compressor + condenser, two pipes.

All-air: single-duct CAV, CAV reheat, multizone, dual duct, VAV. +: central plant, good humidity/filtration, economizer, quiet. −: big ducts, fan energy.

Dual duct CAV: one fan, hot + cold ducts, mixing box per zone; constant air, variable temperature.

CAV vs VAV: fixed air/variable temp vs fixed temp/variable air; VAV saves fan energy.

Rooftop: + packaged, saves floor space, quick; − roof load, weather, service access.

Choice matrix (2024): Σ importance × score: split 346, **rooftop 434**.

VRF

Inverter compressor + EEV per indoor unit, one outdoor unit, many indoor units. +: part-load efficiency, zone control, heat recovery, no ducts, modular, quiet.

3-pipe heat recovery: discharge gas, suction gas, liquid. Heating unit: hot gas in → liquid out. Cooling unit: liquid in → suction gas out. **2-pipe:** 2 pipes to a BC controller that separates gas/liquid.

Multi-split vs VRF: few units, own pipes, same mode, short pipes vs up to 60 units, shared branches, simultaneous heat/cool, long pipes, EEV.

Cooling load

Cooling: unsteady, daytime peak, internal gains added. Heating: steady, pre-dawn peak, no sun or internal gains.

Assumptions: statistical outdoor design data, clear sky, full occupancy, equipment at representative capacity, near-max not max.

```
opaque: U·A·CLTD(ETD)   floor/basement:
U·A·(To-Ti)
glass: A[Ug(To-Ti) + SHGF·SC·CLF]
infil: m = ρ·ACH·Vol/3600; m·cpm·ΔT +
m·2501·ΔW
people: N×S×CLF + N×L ; lights:
W×UF×BF(1.25)×CLF
GTH = RSH + RLH + OATH; ×1.25 safety
```

2024 room: RSH 5035 W, RLH 578 W, ×1.25 = **7.0 kW ≈ 2 TR** (glass U 3.12 assumed, inside 25 °C/50%).

Sol-air temp: $t_{sa} = t_o + \alpha I/h_o - \varepsilon \Delta R/h_o$. **Infiltration:** uncontrolled leakage; crack method = crack length × leakage per m; or ACH method.

Duct design

```
Equal friction: same f (mm H2O/m)
everywhere
D = (0.002268 Q^1.852 / f)^(1/4.973)
De = 1.3(ab)^0.625/(a+b)^0.25 ; 3:1 → b =
0.548D, a = 3b
Fan p = f × longest run + ΣK·pv; pv =
1.2V²/(2×9.81) mm
```

Q m³/s	2	3	5	8	10	14
D @0.1	0.605	0.703	0.850	1.013	1.101	1.248
3:1 a×b	.99×.33	1.16×.39	1.40×.47	1.66×.55	1.81×.60	2.05×.68

@0.09 (2020): 8 → 1.035, 6 → 0.930, 3 → 0.718 m. 2021: 12 → 1.94×0.65, 9 → 1.74×0.58, 4 → 1.29×0.43.

Rules: short and straight, low aspect ratio (≤4:1), velocity limits for noise, gradual transitions, sealed and insulated, dampers for balancing.

Outlet location: throw/spread/drop, near heat sources (windows), no short-circuit to returns, noise, ceiling height. **Dampers** control/balance flow; **diffusers** spread and slow supply air.

Exam plan: In each section answer the 3 questions that contain the numericals above (R22 cycle / ice load in A; summer AC with BPF, cooling load and duct sizing in B) plus definition parts. The numericals repeat almost unchanged.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
ACH	air changes per hour
ADP	apparatus dew point
AHU	air handling unit
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
BPF	bypass factor
CAV	constant air volume
CFC	chlorofluorocarbon
CLF	cooling load factor
CLTD	cooling load temperature difference
COP	coefficient of performance
DBT	dry bulb temperature
DPT	dew point temperature
EER	energy efficiency ratio
ESHF	effective sensible heat factor
ETD	equivalent temperature difference
FCU	fan coil unit
GSHF	grand sensible heat factor
GTH	grand total heat
GWP	global warming potential
HCFC	hydrochlorofluorocarbon
HFC	hydrofluorocarbon
HVAC	heating, ventilation and air-conditioning
IAQ	indoor air quality
IIR	International Institute of Refrigeration
LiBr	lithium bromide
NH3	ammonia
OALH	outside air latent heat
OASH	outside air sensible heat
OATH	outside air total heat
ODP	ozone depletion potential
RE	refrigerating effect
RH	relative humidity
RLH	room latent heat
RSH	room sensible heat
RSHF	room sensible heat factor
RTH	room total heat

SC	shading coefficient
SEER	seasonal energy efficiency ratio
SHGF	solar heat gain factor
TEV	thermostatic expansion valve
TR	ton of refrigeration (3.5 kW)
TSH	total sensible heat
VAR	vapour absorption refrigeration
VAV	variable air volume
VCR	vapour compression refrigeration
VRF	variable refrigerant flow
WBT	wet bulb temperature